Machine Learning Driven Multifactorial Early Dementia Detection: Integrating Genetic and Non-Invasive Clinical Risk Factors

Abstract- A_4_1_40

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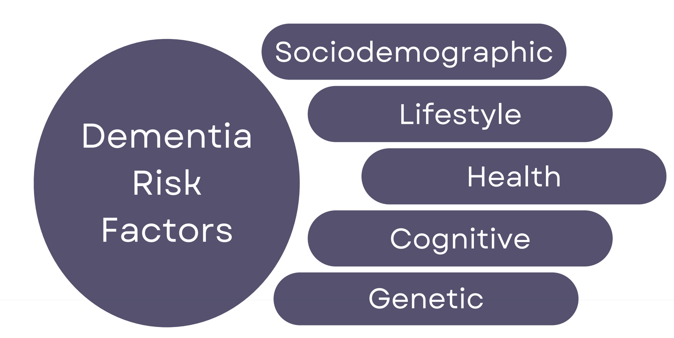
1 INTRODUCTION

- Dementia, a neurological syndrome caused by gradual decline in cognitive abilities with age, is generally difficult to detect , especially in the early stages .
- It has been found to be caused by a combination of lifestyle,medical ,genetic and environmental factors.
- Detection often involves a number of tests including cognitive , clinical assessments and expensive brain imaging tests like MRI , PET scans.
- Invasive methods like Cerebrospinal Fluid Analysis (CSF) are also used in some cases.

This research aims to find an efficient, non invasive machine learning driven alternative for dementia detection by analysing the combined effect of sociodemographic,lifestyle,genetic (APOE4 allele presence), cognitive (MoCA,MMSE scores) and health risk factors identified from the National Alzheimer's Coordinating Center (NACC) dataset.

Dataset:

The National Alzheimer's Coordinating Center (NACC) UDS dataset is a standardized repository of longitudinal clinical, cognitive, genetic, and neuropathological data from participants across 42+ Alzheimer's Disease Research Centers (ADRCs) in the U.S. The dataset contains cognitively normal individuals to those with mild cognitive impairment and dementia.



Initial Dataset Shape:

Rows: 198627

Columns: 1024

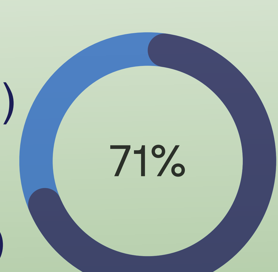
Fig.1 : Dementia risk factor types selected from dataset.

2 METHODOLOGY

Data Cleaning & EDA → **Feature Selection** → **Modelling**

Data Cleaning & EDA

- **D**ropped features with **> 50% missing values** and records with **age(NACCAGE) < 60.**
- Used **Multiple Imputation by Chained Equations (MICE)** to replace NaN values *(better at handling complex variable relationships).*
- Dropped low variance features using Variance Threshold. **No. of filtered features= 64.**
- Split dataset into train,validation ,test sets. Split ratio - 70:15:15
- Initial class imbalance (Training set):
  0 91321
  1 37472



- Applied **SMOTE** on training set to generate minority class (Demented) samples. Class Distribution After SMOTE:
  0 91321
  1 91321

Feature Selection

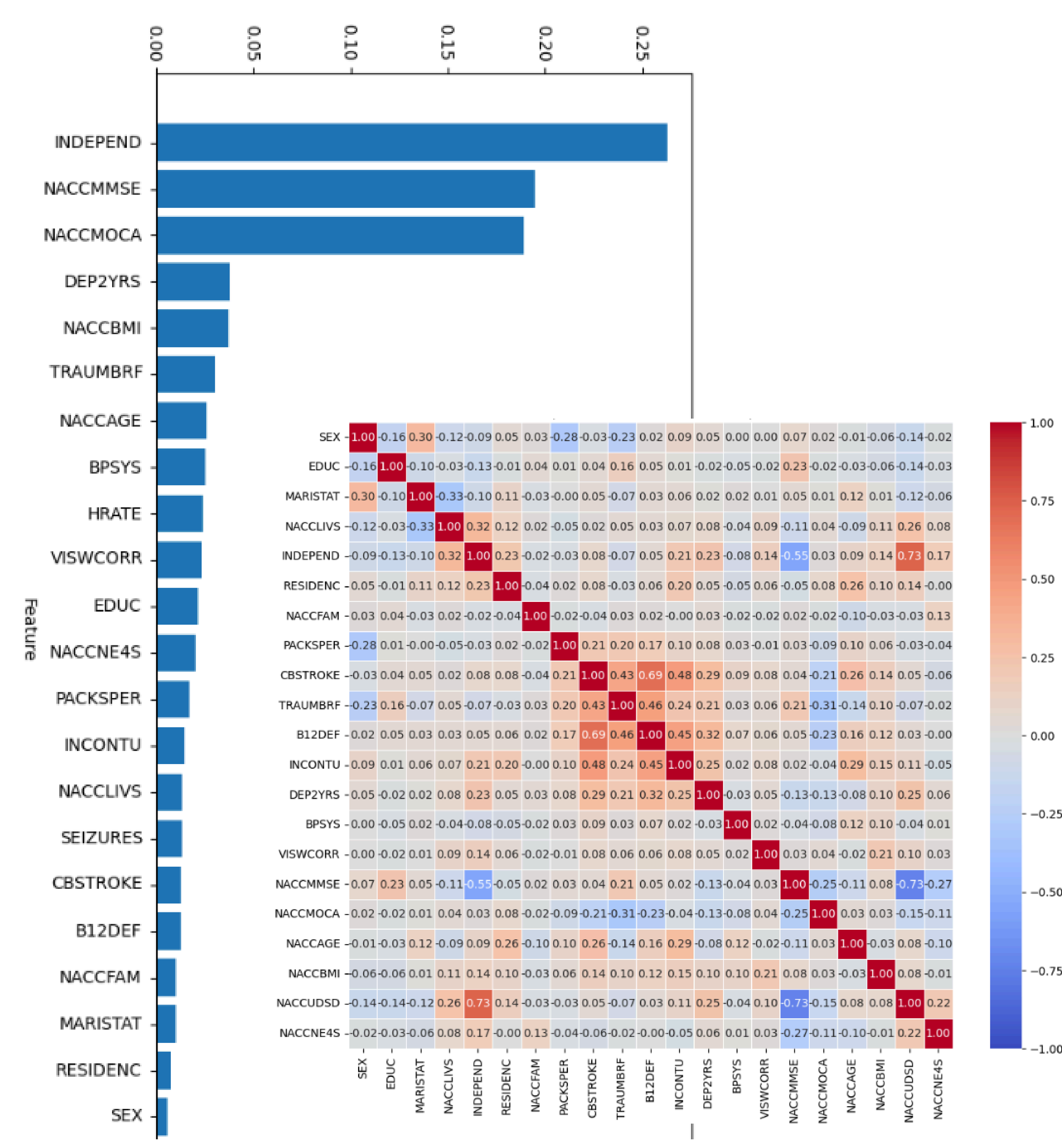


Fig.2 : Feature Importance Graph , Feature Correlation Matrix

Modelling

- Selected best feature set by comparing **Random Forest Feature Importance, Recursive Feature Elimination (RFE)** values applied on cleaned feature set, also validated feature ranking against existing Dementia literature.
- Derived Pearson ,Spearman Correlation matrix for the filtered feature list to check for features with high collinearity. These were further removed to reduce multicollinearity.
  Final number of features = 21
- Trained **Random Forest** on training set with k-fold cross validation .
  Model selection was based on:
  1. Scalability for large datasets.
  2. Ability to handle mixed datatypes ,robustness to outliers.
  3. Level of interpretability.
- Finally,**Randomized Search CV** was used to optimize hyperparameters for the model to obtain best test set performance.

3 RESULTS AND DISCUSSION

Test-Set Performance

		precision	recall	f1-score
NOT DEMENTED	0	0.95	0.94	0.94
DEMENTED	1	0.86	0.87	0.86

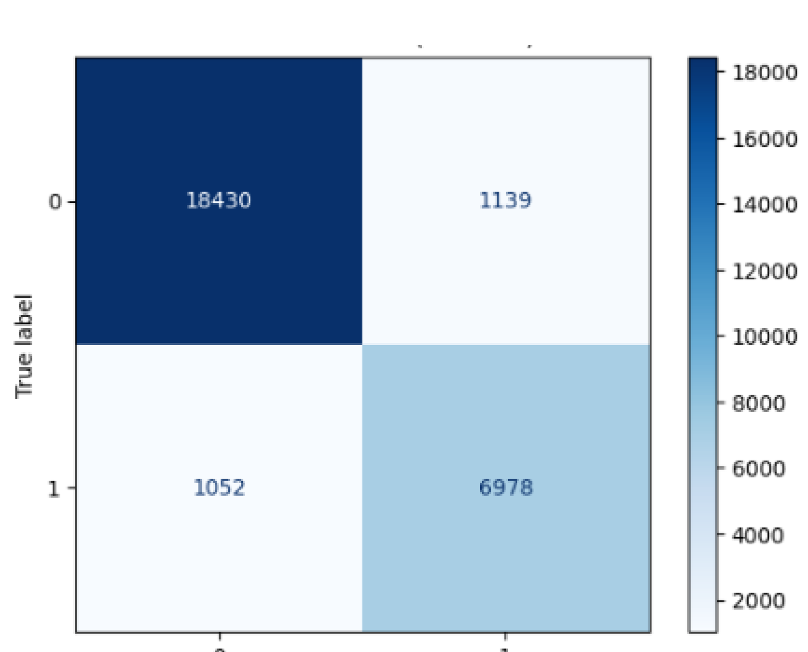


Fig.3 : Confusion Matrix (Test Set)

Accuracy: 92.06%

ROC-AUC Score: 96.55%

Top contributing feature:

INDEPEND (Level of Independence)

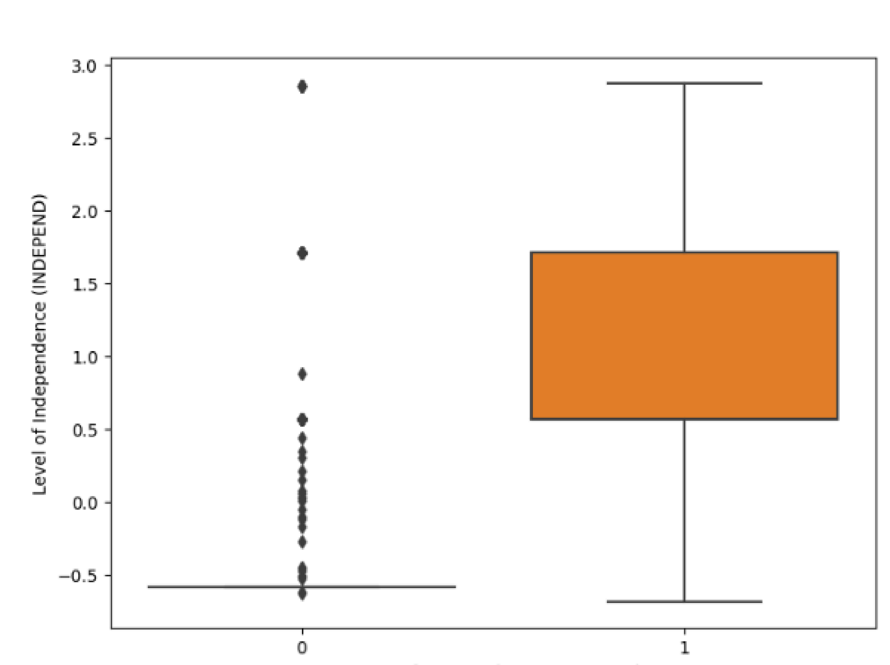


Fig.4 : INDEPEND by DEMENTIA status (Test Set)

On average, **DEMENTED** class showed lower independence but with greater spread in values .

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We hereby declare that the submitted research poster is original and has not been presented or published elsewhere.